

INTPART summer school, 2025

Your name goes here

December 17, 2025

Abstract

A short overview of what your report is about.

1 Introduction

2 Nuclear physics and nuclear astrophysics

1. How you add bullets
2. Add a new bullet

2.1 Elements and their production in the Solar System

2.2 Nuclear Burning stages of stars

2.3 The production of heavy nuclei

s-process

r-process

p-process

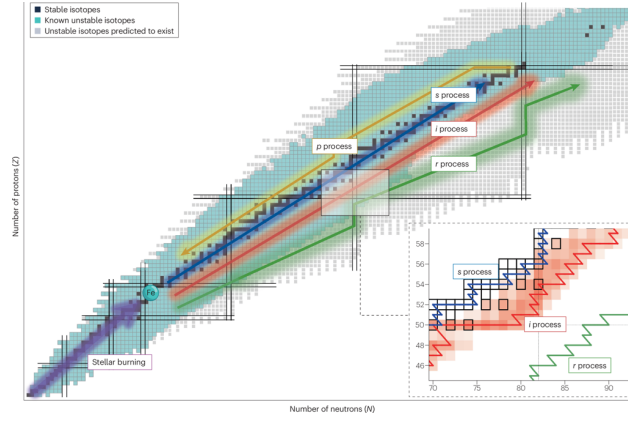


Figure 1: Diagram showing the nuclide chart with the different nucleosynthesis sites. [1]

Table 1: Table showing the typical energy ranges for the different resonances

resonance	Typical energy range (MeV)
GDR	

3 Nuclear Level Density and Gamma Strength Function

4 TALYS

5 TALYS exercises

The following TALYS simulations were performed using TALYS version 1.96 [2].

5.1 Exercise 1

5.1.1 ^{191}Os

(n, γ)

(n, n')

5.1.2 ^{191}Re

(n, γ)

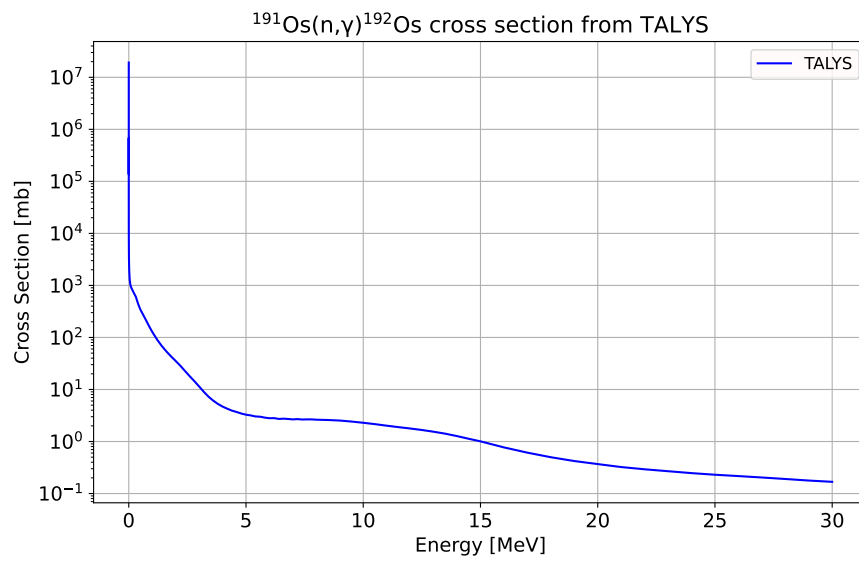


Figure 2: Caption

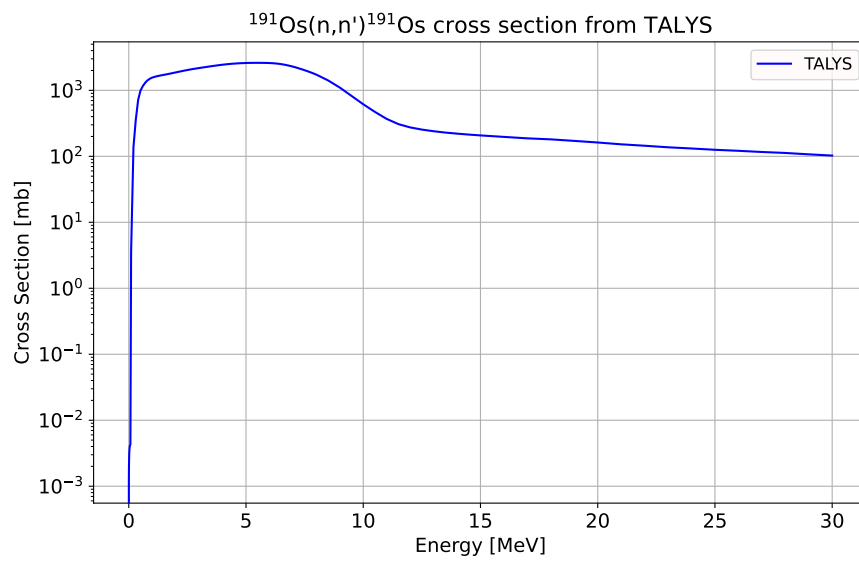


Figure 3: Caption

5.2 Exercise 2

5.2.1 ^{208}Pb

(n, n')

$(n, 2n)$

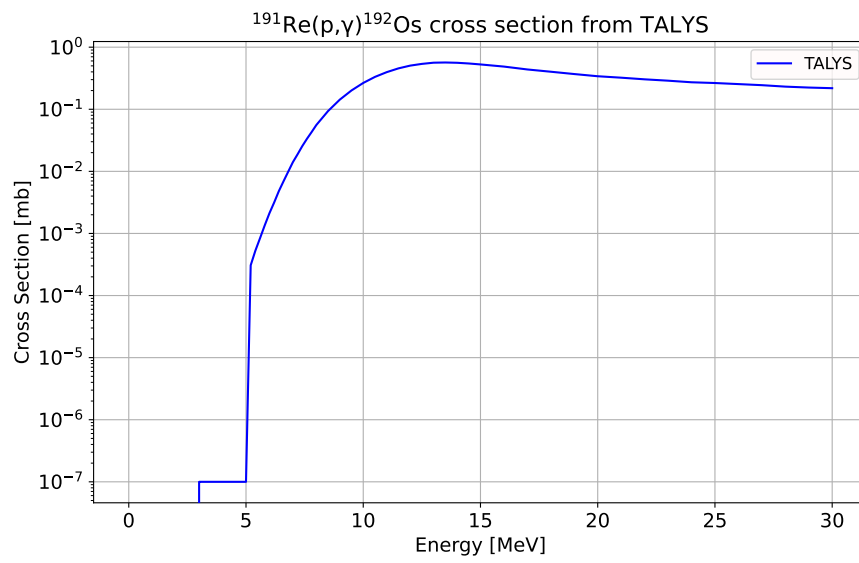


Figure 4: Caption

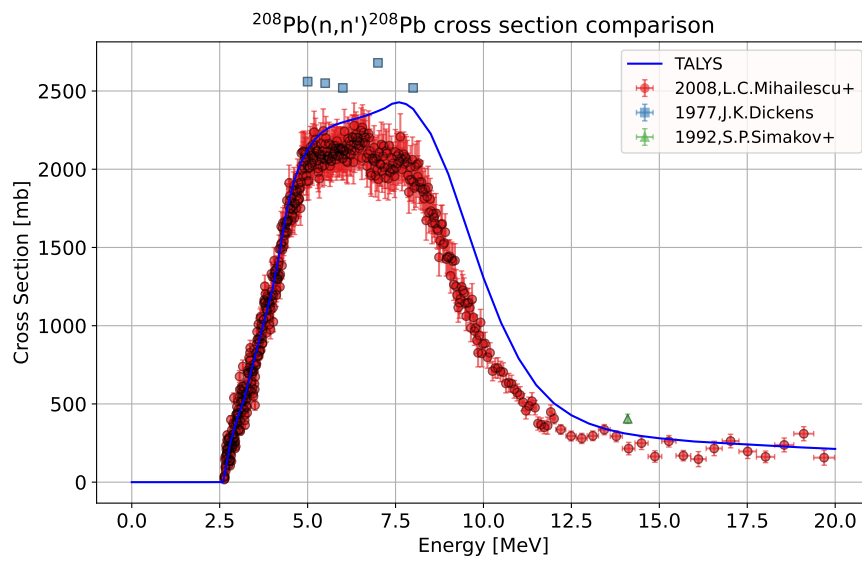


Figure 5: Caption

(n, γ)

5.3 Exercise 3

5.3.1 ^{208}Pb With Varying Models

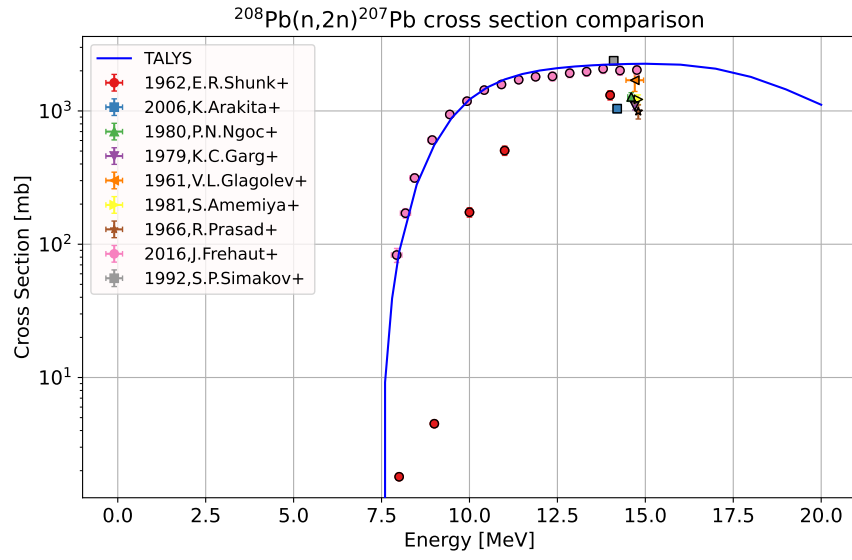


Figure 6: Caption

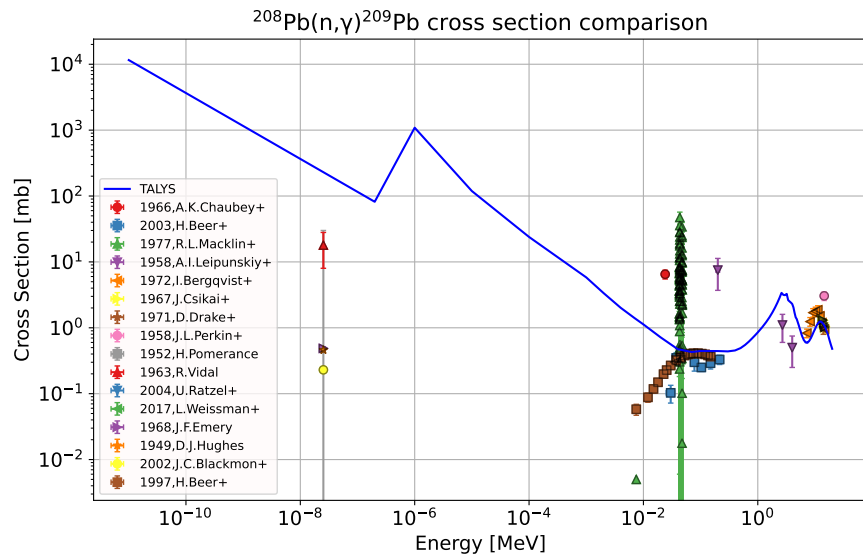


Figure 7: Caption

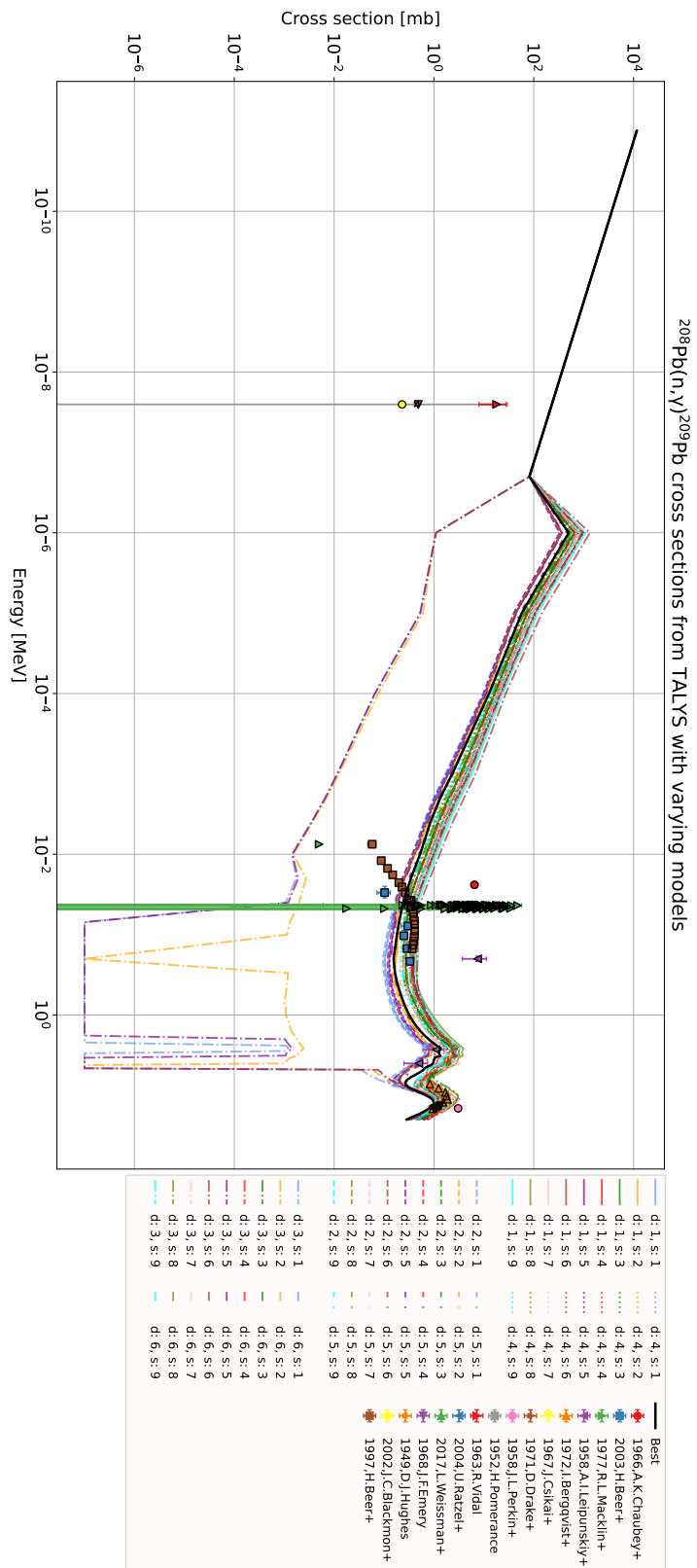


Figure 8: Caption

5.4 Exercise 4

5.4.1 $^{191}\text{Os}(n, \gamma)$ With Experimental γ -Ray Strength Function

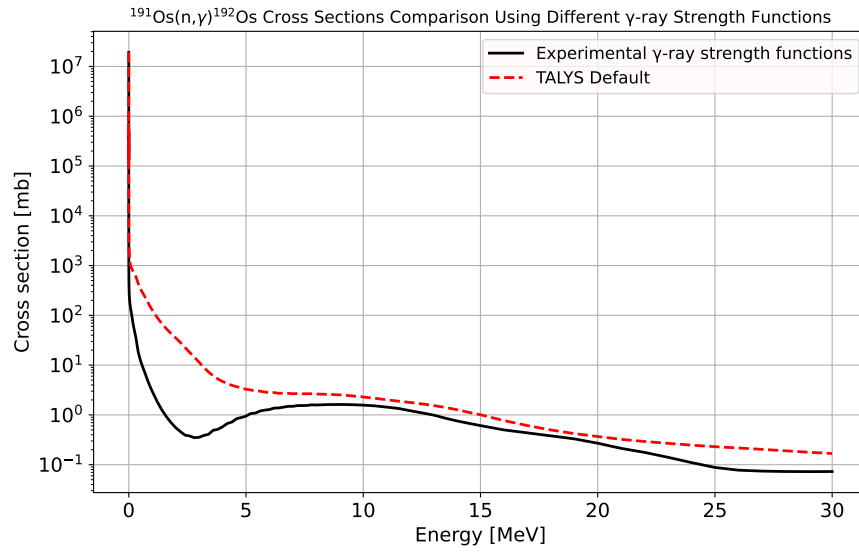


Figure 9: Caption

5.5 Task

5.5.1 $^{120}\text{Sn}(n, \gamma)$

test

6 Conclusion and Outlook

Appendix A:

References

- [1] Mathis Wiedeking et al. *Unlocking i-process nucleosynthesis by bridging stellar and nuclear physics*. 2025. DOI: [doi:10.1038/s42254-025-00885-7](https://doi.org/10.1038/s42254-025-00885-7). URL: <https://doi.org/10.1038/s42254-025-00885-7>.
- [2] Arjan Koning, Stephane Hilaire, and Stephane Goriely. “TALYS: modeling of nuclear reactions”. In: *The European Physical Journal A* 59.6 (June 2023), p. 131. ISSN: 1434-601X. DOI: [10.1140/epja/s10050-023-01034-3](https://doi.org/10.1140/epja/s10050-023-01034-3). URL: <https://doi.org/10.1140/epja/s10050-023-01034-3>.